

This topic will begin with an introduction to the first tools in analytics and analytics dashboard. You will then learn about the dimensional model and its types and its application in business analytics. To adopt and practise business analytics and to become a good analyst, these are concepts that you are expected to be aware of.

It is important to note that this week requires some amount of self-study to further develop these skills.

But before you begin, participate in the discussion below and share your responses with your classmates.



Activity

Analytics tools

Using the padlet below, conduct a web search and share examples of some analytics tools that you are aware of and that you may have used in your job.

Include the following:

- If you have used a particular tool, what are its best features?
- What else would you like the tool to do for you? In other words, what improvements would you recommend?

To use the padlet

Double click anywhere on the board or tap/click on the pink + to add to the padlet. If you want to open the padlet fullscreen, click on the 'open in new' icon in the top right of the padlet screen. Get creative with photos, videos, LinkedIn profile URLs and more.

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2025 BUS5PB S1: Week 3: Topic 1 - Analytics tools

Using the padlet below, conduct a web search and share examples of some analytics tools that you are aware of and that you may have used.



Made with **Padlet**

Introduction to first tools in analytics

The first tools in analytics are almost always focused on data visualisation and the integration of such visualisations into interactive dashboards. Some of the most commonly used tools are:

- Microsoft Power BI
- SAP Lumira
- Tableau
- QlikView
- Yellowfin BI

Microsoft Excel is often used for visualisation too. It not only allows you to collect and view data, it can also be used to plot graphs.

As an analyst, some of the main features that an analytics tool must possess are:

- ability to import data from multiple sources
- offers a wide variety of charts
- easy to use
- allows integration with other advanced tools
- portability of dashboards.

Analytics dashboards

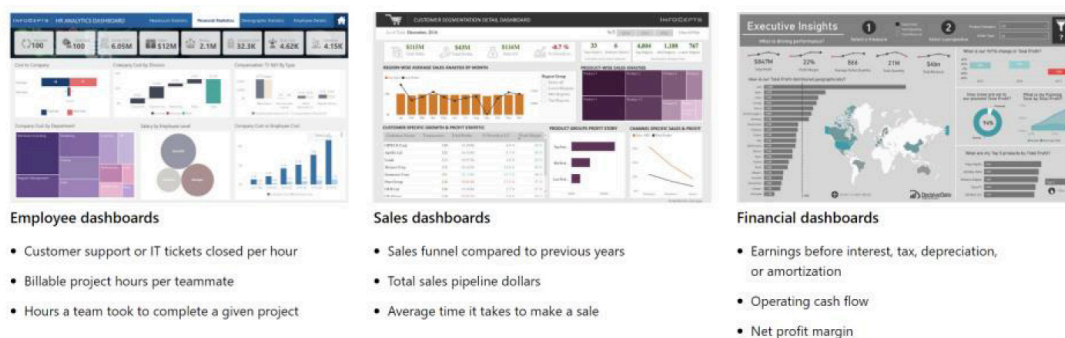
An analytics dashboard is a collection of visual (and text) widgets that represents all the data, insights and interaction on a single screen. Dashboards can also connect all types, formats and sources of data with different metrics and key performance indicators (KPIs).

A typical dashboard will always start with summary information to provide a broad view of the overall numbers and then expand into each category in detail.

Analytics dashboards are used across all departments within an organisation, as shown in the image below. For example, an employee dashboard will be used by the human resources team, a sales dashboard will be used by the sales and marketing team and a financial dashboard will be used by the finance team. These dashboards often include common metrics or KPIs. For instance, the number of billable hours of an employee will reflect the sales generated by the employee and determine the contribution to the company's finances. It is important to note that these dashboards are often used by all employees, unlike in the past where only the C suite executive used them to track performance.

Figure 3.1

Data dashboards



(Click on the image to expand)

© [Microsoft Power BI](#).

[\[Image transcript\]](#)

Let's explore one of the most commonly used tools, Microsoft Power BI, in detail.

Microsoft Power BI

Gartner has recognised Microsoft Power BI in the Leader Quadrant of 'Analytics and Business Intelligence Platforms' for 14 consecutive years, including 2021.



Read

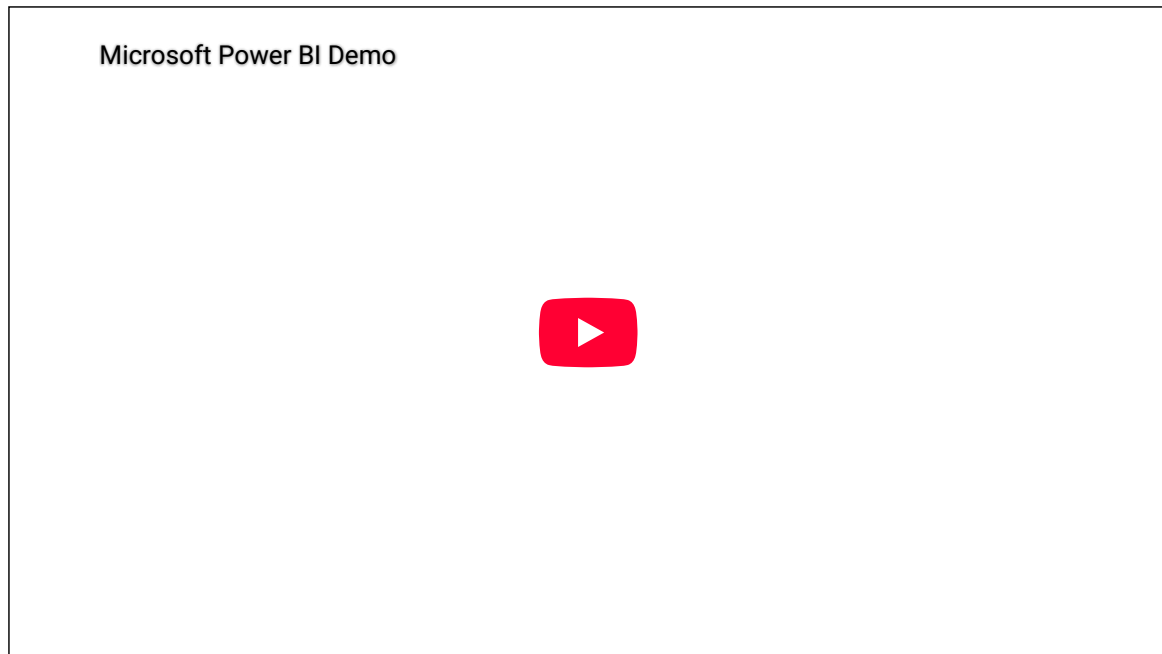
At this website, you will learn more about [Microsoft Power BI](#) and why it is the most appreciated tool in business analytics. As an analyst, you may be expected to use this tool in your day-to-day activities. Therefore, having the knowledge of this tool and how it works is critical.



Watch

Microsoft Power BI demo

This video [11.12 min.] is a basic demonstration of Microsoft Power BI and introduces some of its functionalities. Watch this video to get familiar with the tool. This will be required to complete the activities in the workshop at the end of the week and also for Assessment 2.



© 2019 by Solution Systems, Inc. *Microsoft Power BI Demo*

Dashboard best practices

There are many guidelines and best practices available from different providers and vendors, but some of the best practices to follow are:

- identify and develop a well-defined objective driving the dashboard
- use the most up-to-date data
- group data logically
- know your graphs and data type (line graph — temporal data)
- know your audience —technical/non-technical
- maintain a hierarchy of information and insights
- maintain consistent interface design
- ensure consistent use of colour.

There are many more than those listed here.

Dashboard practices to avoid

Some of the practices to avoid are:

- the negation of the 'Follow' list
- too many widgets, cluttered visuals
- incorrect or low-quality data
- poorly formed graphs — missing labels, axes
- trying to reach multiple audiences
- excess animation or expectation of user interaction.

Again, there are many more than listed above.



Read

Visit [this page](#) to explore best Power BI dashboard examples.

A first data model

A fairly simple and intuitive data model underlies all of the analytics tools discussed thus far. It's called the dimensional model. The dimensional model originated in the data warehousing era. Formally defined, a dimensional model is a database structure that is optimised for online queries and data warehousing tools. A first data model comprises:

- measures or facts
- dimension or descriptors.

Its advantages include:

- being intuitive to 'business' users
- fast query performance
- easily extensible.

So, what about the design of data representation in the data warehouse? When designing a data warehouse (or a data mart), it is based on the concept of dimensional modelling for speed of access. The rules of normalisation are replaced by a data model that is organised around the concepts of 'facts' and 'dimensions'. The two (common) database design schemas are known as:

- stars, and
- snowflakes.

Measures or facts

Measures or facts are a numeric variable expressing the KPIs, performance measurements or any other quantifiable information about an organisation. The information represented by 'measures' is used to support or evaluate the decision making and performance of an organisation. The tables that hold measure information are known as the fact tables. The numeric measures in a fact table fall into three categories, being additive, semi-additive and non-additive, and can have both discrete and continuous values.

Click on the below headings to reveal further information.

▼ Additive

The most flexible and useful facts are fully additive. Additive measures can be summed across any of the dimensions associated with the fact table, e.g. sales amount.

> Semi-additive

> Non-additive

Dimensions or descriptors

Dimensions describe the 'who, what, where, when, how and why' associated with the measures.

For example, 'Customer — name, dob, address, preferences'.

Dimensions are categorical variables, and take nominal, ordinal, binary or text types of values. A dimension is always used to 'describe' (or spread out) an aggregated measure. A dimension 'table' is a grouping of a similar set of dimensions (or attributes or columns), like the 'customer table'.

A hierarchy is a structure made up of two or more levels of related dimensions. A dimension at an upper level of the hierarchy completely contains one or more dimensions from the next lower level of the hierarchy. For example, the 'State' dimension would contain one or more 'Wine region' dimension in the example given later in this topic. Hierarchies enable us to navigate to different levels of 'granularity' within the measures; that is, we can look at the measures at one level of the hierarchy and then drill into a selected member to see the next lower level. We will explore the Wine region example in detail, but before that let's look at the key elements of a dimensional model.

Key elements of a dimensional model

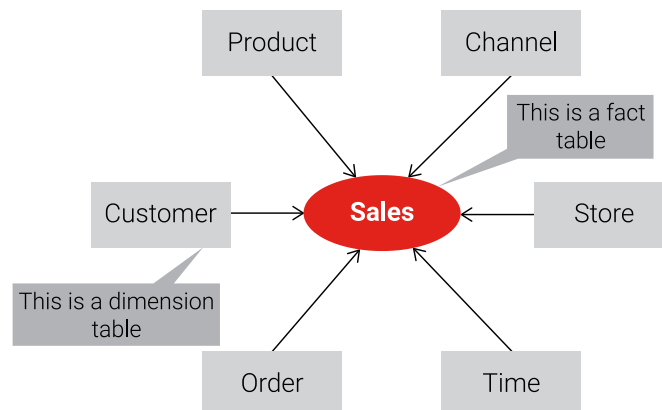
The key elements of dimensional modelling are:

- measures (or facts) — e.g. sales performance
- dimensions — e.g. time, product, market and region
- keys — surrogate key, foreign key, composite key
- attributes, hierarchies, schema.

In a dimensional model, each 'dimension' table has a primary key and its own attributes. Surrogate keys are commonly used as primary keys for dimension tables in data warehouse designs. In a 'fact' table, the primary key is almost always defined as a subset of the foreign keys supplied by the dimensions. In most environments, this composite key will suffice as the primary key to the fact table.

Figure 3.2

Dimensional modelling



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[\[Image transcript\]](#)

When a dataset (CSV file or database table) is imported into an analytics tool, it automatically creates a dimensional model based on the types and occurrences of the data points in this dataset. When you click and drag fields on to the dashboarding canvas, you would be mainly visualising a numerical field with a categorical field, such as:

- sales revenue by month
- customer count by suburb
- expenses by expense type.

When you are thinking in dimensions, the key points to remember are:

- measures are quantitative data items that give a numeric value to the data (sentiment, count, revenue, as shown in the example below)
- dimensions add context to the measures and without context, measures are meaningless numbers (age group, product type, competitor, region, country, as shown in the example below).

Figure 3.3

Measures example

	A	B	C	D	E	F	G
1	Age Group	Product Type	Competitor	Region	Country	Customers Measures	Value
2	12 to 18	Educational	Lumo	Americas	Mexico	Sentiments	6.525701459
3	12 to 18	Educational	Lumo	Americas	Mexico	Sentiments	2
4	12 to 18	Educational	Lumo	Americas	United States	Sentiments	9.85050505
5	12 to 18	Educational	Lumo	Americas	United States	Sentiments	2
6	12 to 18	Educational	Lumo	Americas	United States	Count	61.6
7	12 to 18	Educational	Lumo	Americas	United States	Count	2
8	12 to 18	Educational	Lumo	Americas	United States	Revenue	85.5
9	12 to 18	Educational	Lumo	Americas	United States	Revenue	2
10	12 to 18	Educational	Lumo	Americas	Canada	Count	15
11	12 to 18	Educational	Lumo	Americas	Canada	Count	2
12	12 to 18	Educational	Lumo	Asia Pacific	Singapore	Count	16

(Click on the image to expand)

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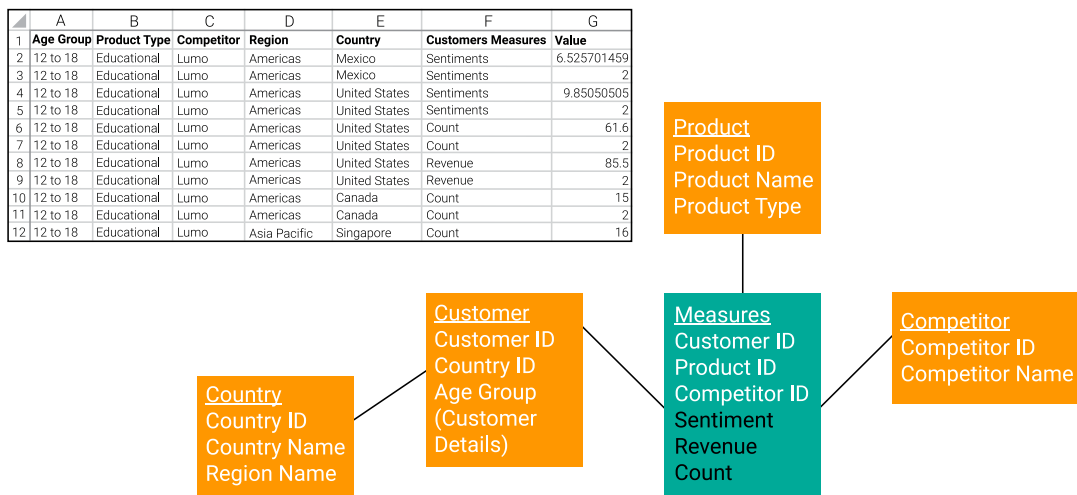
[\[Image transcript\]](#)

Dimensional model — examples

A typical dimensional model is represented as shown in the image below.

Figure 3.4

Dimensional model example



(Click on the image to expand)

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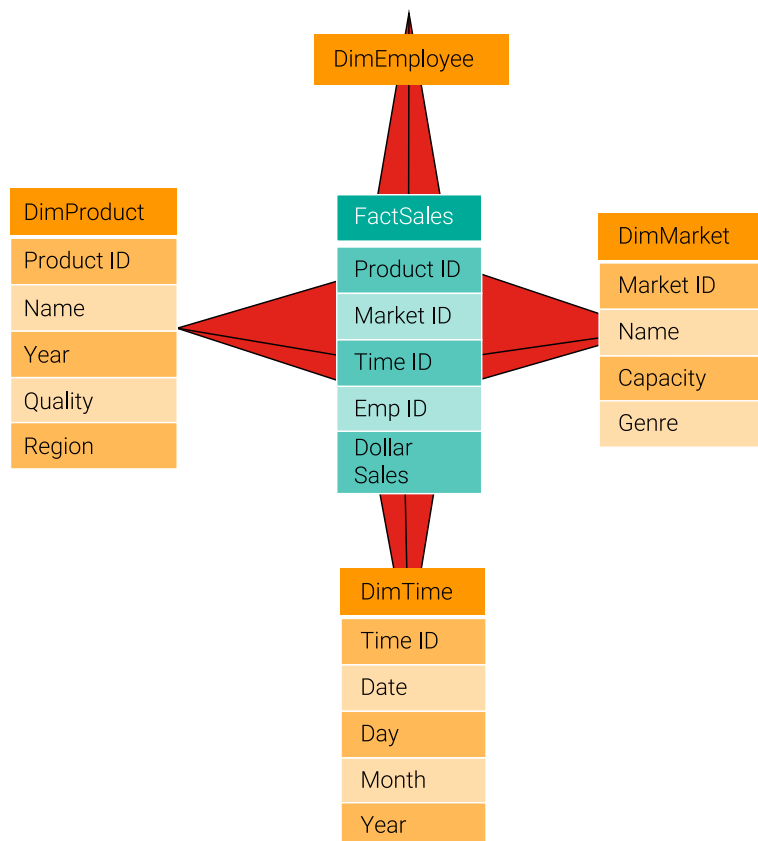
[\[Image transcript\]](#)

• Star schema

- The star schema uses two types of tables — fact tables and dimension tables — and brings the elements together in the shape of a star as shown in the image below, hence the name star schema.
- The centre of the star is formed by the fact table. The fact table has a column for the measure (fact) and a column for each dimension containing the foreign key for a member of that dimension.
- The points and edges are dimension tables that contain the descriptive context of these measures.
- The primary key for the fact table is created by concatenating all of the foreign key fields known as the composite key.
- The dimension table has a column for the unique identifier of each member of the dimension known as an integer or a short character.
- The dimension table has additional columns for attributes that describe/define each member of the dimension.

Figure 3.5

Star schema example

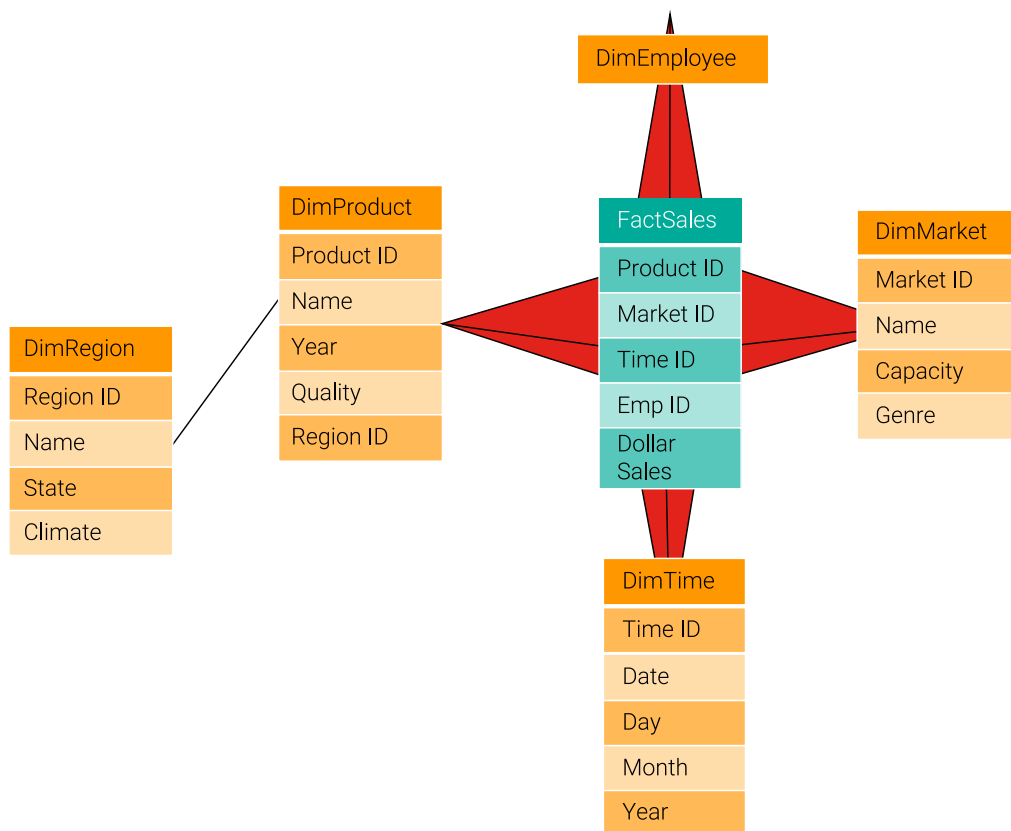


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[\[Image transcript\]](#)

- **Snowflake schema**

- A snowflake schema would contain a normalised dimension table to separately represent a hierarchy.
- Each level of hierarchy is stored in a separate dimension table and joined to its parent dimension (the upper level) as shown in the image below.

Figure 3.6*Snowflake schema example*

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[\[Image transcript\]](#)

Let's explore the dimensional model using an example of a data set about wine regions and wine production.

Example 1: One dimension

A company says: 'We sell wine from different regions in various markets and measure our sales performance over time.'

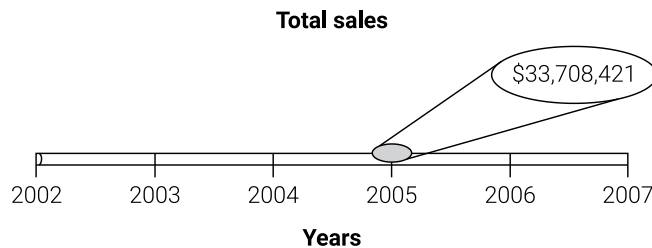
For this statement, you need to measure total sales performance as shown in the image below.

Figure 3.7*Example 1a*

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Now, if you introduce a time dimension, the model will be represented as below:

Figure 3.8*Example 1b*



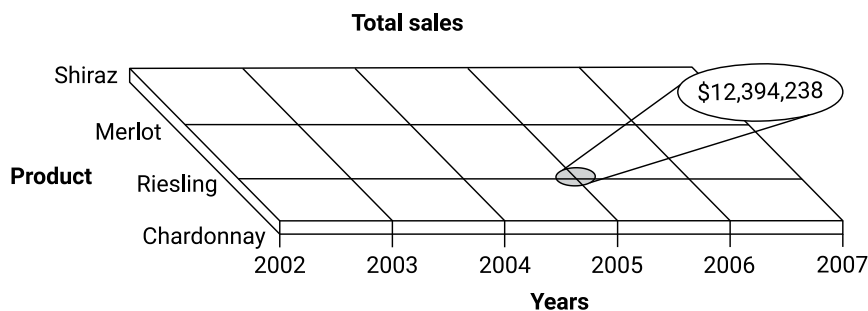
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Example 2: Two dimensions

The same company says: 'We sell wine from different regions in various markets and measure our sales performance over time.'

In such a case, you will introduce a product (wine) dimension and the model will be represented as shown below:

Figure 3.9
Example 2



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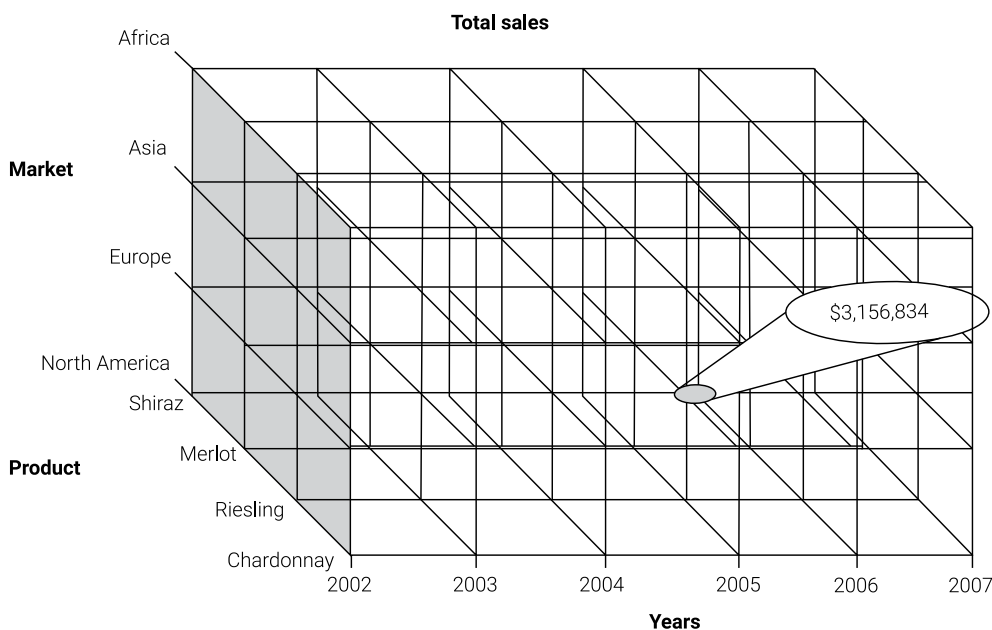
[\[Image transcript\]](#)

Example 3: Three dimensions

The same company says: 'We sell wine from different regions in various markets and measure our sales performance over time.'

In such a case, you will introduce a market dimension and the model will be represented as shown below:

Figure 3.10
Example 3



(Click on the image to expand)

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[\[Image transcript\]](#)

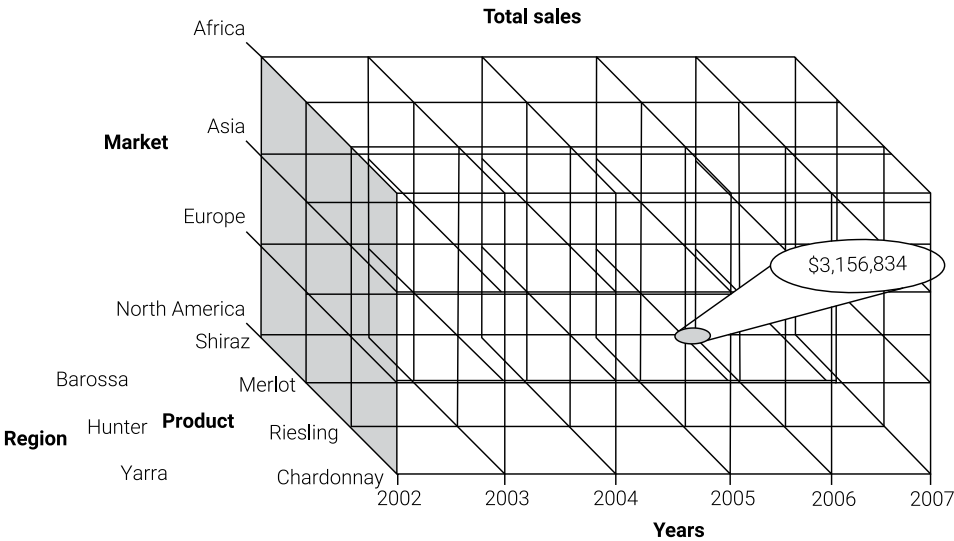
Example 4: Four dimensions

The same company says: ‘We sell **wine** from different regions in various markets and measure our sales performance over time.’

In such a case, wine region can be a dimension or a dimensional attribute and is not directly linked to sales performance as shown in the image below.

Please note that more than three dimensions is difficult to visualise but you can keep adding.

Figure 3.11
Example 4



(Click on the image to expand)

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[\[Image transcript\]](#)

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